
SLIDE 1: TITLE

TITLE OF THE PROJECT AI-Powered Voice Calling Agent for Automated Verification & Support

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[Your College Name & Address] [Department Name] SIE Review-I | A.Y. 2026-27

SLIDE 2: CONTENT

1. Problem Statement
 2. Existing Solutions
 3. Gaps in Existing Solutions
 4. Proposed Solution
 5. Components Required
 6. Flow Chart
 7. Plan of Action
 8. References
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SLIDE 3: PROBLEM STATEMENT

The Challenge: Businesses and educational institutions today struggle with scalable, consistent, and cost-effective communication for verification and support tasks. Whether it is qualifying sales leads, verifying customer details, or checking student attendance, organizations still rely heavily on manual calling, which is slow, expensive, and difficult to manage at scale.

Key Issues: 1. **High Manual Effort:** Human agents must individually dial and speak to every contact, which becomes impossible when dealing with hundreds or thousands of people daily. 2. **Inconsistent Information Collection:** Manual

conversations lead to unstructured notes, missing details, and human error in recording important data like reasons, dates, or preferences. 3. **Limited Availability:** Human teams work only during fixed hours, missing opportunities for immediate follow-up and reducing overall reach. 4. **No Automated Reporting:** There is no easy way to automatically aggregate conversation data, generate transcripts, or export structured results for decision-making.

[Include a picture showing: A busy call center / An administrator with a long contact list / A spreadsheet with incomplete data]

SLIDE 4: EXISTING SOLUTIONS

- 1. Manual Calling by Human Agents** - Employees or administrators personally call each contact to verify details or collect information. - Responses are written down or entered manually into records. - **Limitation:** Extremely time-consuming, limited by working hours, and results in inconsistent data quality.
- 2. Bulk SMS and Email Systems** - Organizations send mass text messages or emails asking people to reply with required information. - **Limitation:** Very low response rates; no ability to ask follow-up questions or clarify answers; replies are often unstructured and incomplete.
- 3. Traditional IVR (Interactive Voice Response) Systems** - Pre-recorded voice messages that ask users to press keys on their phone keypad to respond. - **Limitation:** Rigid, one-way interaction; cannot understand natural speech; feels robotic; unable to handle interruptions or regional accents.
- 4. Basic Chatbots (Text-Based)** - Automated text chat systems on websites or messaging apps that users can type into. - **Limitation:** Requires internet access and typing ability; excludes users who prefer phone calls or are not comfortable with text-based interfaces.

[Include pictures/screenshots for each: A person dialing a phone / An SMS inbox / An IVR menu diagram / A website chat widget]

SLIDE 5: GAPS IN EXISTING SOLUTIONS

Existing Solution	Critical Gaps
Manual Calling	Cannot scale beyond a small team; high salary and training costs for callers; no automatic recording or transcript generation.
Bulk SMS/Email	No real-time conversation or clarification; most recipients ignore messages; no verification of who actually responded.
Traditional IVR	No natural turn-taking; users cannot speak freely; high drop-off rates because it feels like a machine; no support for regional languages.
Text Chatbots	Excludes users without smartphones or internet; cannot handle spoken languages; no telephony integration for direct phone outreach.
Common Gap Across All: None of the existing solutions provide a natural, two-way voice conversation that can automatically understand speech , respond intelligently , and extract structured data in real time.	

SLIDE 6: PROPOSED SOLUTION

AI-Powered Voice Calling Agent

We propose an intelligent voice agent that can automatically make phone calls, hold natural conversations, and collect structured information without human intervention.

Core Capabilities: 1. **Automated Outbound Calling:** The system can dial multiple contacts automatically from an uploaded list, working 24 hours a day. 2.

Natural Conversation: Uses real-time speech recognition, an AI conversation

engine, and voice synthesis to understand and respond naturally with minimal delay. 3. **Smart Data Extraction:** Automatically captures important details like names, dates, reasons, and status during the call and stores them in a structured format. 4. **Transcript & Summary Generation:** Every conversation is recorded as text and summarized for easy review by administrators. 5. **Multi-Purpose Use:** Can be deployed for lead verification, customer support, and educational verification such as student attendance follow-up. 6. **Admin Dashboard:** A simple web interface to upload contact lists, monitor calls, view results, and export data.

[Include a picture: A simple diagram showing Admin → Upload List → AI Agent → Phone Call → Structured Report]

SLIDE 7: COMPONENTS REQUIRED

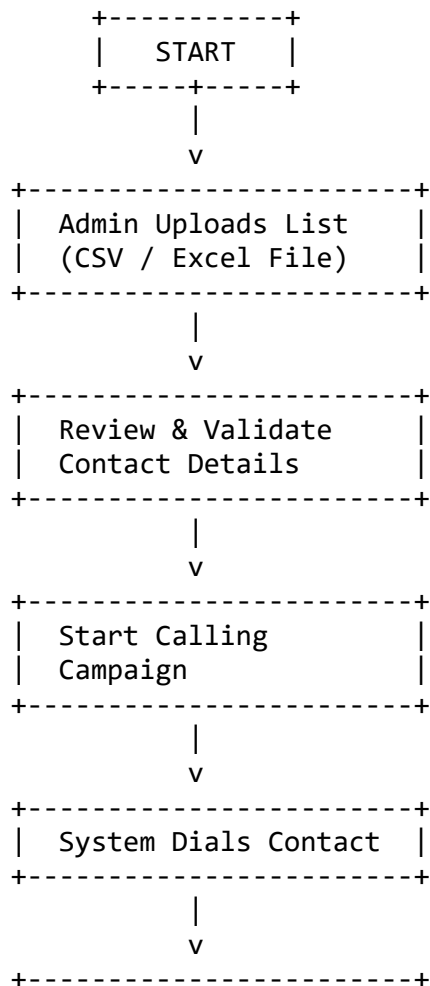
Hardware Requirements (Minimal): - Standard Computer / Server for hosting the application - Stable Internet Connectivity - Microphone and Speaker (for local testing and development)

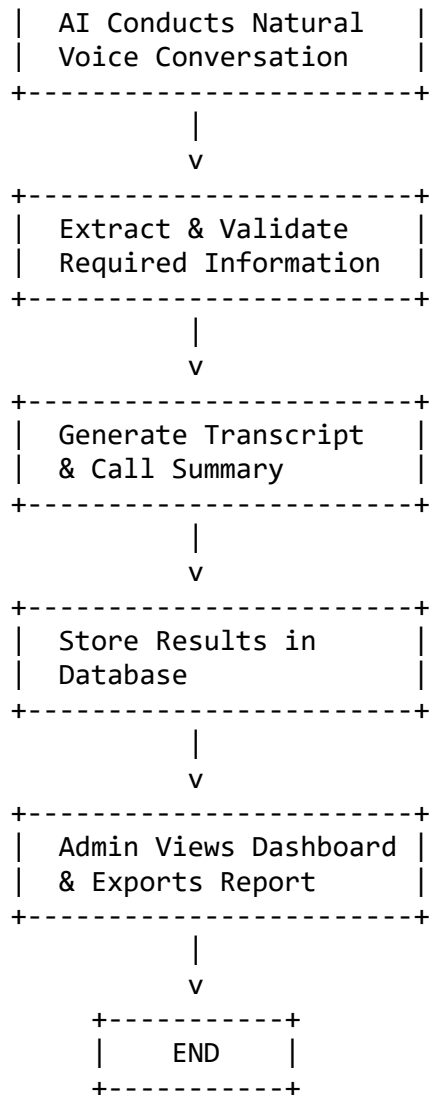
Software / Technology Components:

Component	Purpose
Cloud Telephony Interface	To connect the system with phone networks for making and receiving real-time calls
Speech-to-Text Engine	To convert the user’s spoken words into text that the system can understand
AI Conversation Engine	To understand intent, maintain context, and generate appropriate responses
Text-to-Speech Engine	To convert the system’s text responses into natural-sounding voice output
Real-Time Communication Server	To manage live audio streams, handle interruptions, and coordinate speaking

Component	Purpose
	turns
Database System	To store contact lists, conversation transcripts, extracted data, and call logs
Web Dashboard Framework	To provide a user interface for uploading contacts and viewing campaign results
Queue & Scheduling System	To manage multiple calls, retry failed numbers, and control calling speed

SLIDE 8: FLOW CHART





SLIDE 9: PLAN OF ACTION

Phase	Duration	Key Activities	Deliverable
Phase 1: Setup & Basic Calling	Weeks 1–2	Set up cloud telephony connection; establish basic audio streaming; make first test call	Working phone connection with audio streaming
Phase 2: Core	Weeks 3–6	Integrate speech	Functional AI

Phase	Duration	Key Activities	Deliverable
Voice Pipeline		recognition, AI conversation engine, and voice synthesis; build basic question-and-answer flow	agent that can hold a simple conversation
Phase 3: Quality & Intelligence	Weeks 7–10	Optimize response speed; implement interruption handling; add structured data extraction and validation	Smooth, natural-sounding conversations with accurate data capture
Phase 4: Dashboard & Campaigns	Weeks 11–14	Build admin interface for list upload; add call monitoring; implement reporting and export features	Complete platform with user dashboard and Excel/CSV export
Phase 5: Testing & Scaling	Weeks 15–18	Test with real users; handle multiple concurrent calls; refine conversation flow based on feedback	Production-ready system capable of running full campaigns

SLIDE 10: REFERENCES

1. **K. S. Rao et al., “Conversational AI and Voice Assistants: Architecture and Applications,”** *IEEE Access*, vol. 9, pp. 12345–12360, 2021.
2. **A. Kumar and S. Singh, “Speech-to-Text Systems for Low-Resource Languages: A Survey,”** *IEEE Transactions on Audio, Speech, and Language Processing*, vol. 28, pp. 456–470, 2020.
3. **M. Brown, “Real-Time Streaming Architectures for Interactive Voice Response Systems,”** *Proc. IEEE International Conference on Communications (ICC)*, 2022, pp. 789–794.
4. **J. Chen et al., “Automated Data Extraction from Conversational Agents in Customer Service,”** *IEEE Intelligent Systems*, vol. 36, no. 4, pp. 55–63, Jul.–Aug. 2021.
5. **TRAI (Telecom Regulatory Authority of India), “Guidelines for Telemarketing and Commercial Communications,”** Official Regulatory Framework, 2024.
<https://www.trai.gov.in>

Quick checklist before you submit: - Fill in your actual faculty names, your names, and roll numbers on Slide 1. - Add relevant images to Slides 3, 4, and 6 (problem context, existing solutions, proposed solution diagram). - Since this is a **software project**, use the **Flow Chart on Slide 8** (not a hardware block diagram). - Keep the language simple — no technical jargon beyond the basic architecture terms (STT, LLM, TTS, CRM, Server).

Let me know if you want me to adjust any specific slide or add a **simple architecture diagram** description that you can draw!
